MIL-D-17650A(SHIPS)
20 April 1966
SUPERSEDING
MIL-D-17650(SHIPS)
21 August 1953
(See 6.2)

MILITARY SPECIFICATION

DISK, VALVE, RUBBER; INSERT TYPE

1. SCOPE

1.1 This specification covers insert type rubber valve disks for sprinkler, void tank flooding in sea water service.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Setup.

MILITARY
MIL-P-116 - Preservation, Methods of.

STANDARDS

FEDERAL
FED-STD-601 - Rubber: Sampling and Testing.

MILITARY
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

3. REQUIREMENTS

3.1 Material. - The valve disks shall be made of synthetic rubber in sizes as specified (see 6.1).

3.2 Tolerances. - The tolerances allowed on the valve disks shall be as specified in table I.

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Table I - Tolerances.

	Tolerances in inches	
	Plus	Minus
Outside diameter	0	1/32
Inside diameter	1/32	0
Thickness	1/32	1/32

3.3 Physical requirements. - The synthetic rubber in the valve disk shall conform to the physical requirements specified in table II, when tested as specified in 4.4.

Table II - Physical requirements of synthetic rubber.

Properties	Requirement	Test Method
Initial properties:		
Tensile strength, psi, minimum	2000	4.4.2
Ultimate elongation, percent, minimum	75	4.4.2
Hardness, durometer points	85 ± 5	4.4.3
Properties after oven aging for 70 ± 1/2 hours at 100° ± 1.1°C. (212° ± 2°F.):	----	4.4.4
Tensile strength, percent of initial, minimum	80	4.4.4.1
Ultimate elongation, percent of initial, minimum	70	4.4.4.1
Hardness change, durometer points, maximum	5	4.4.4.2
Compression set, percent, maximum	65	4.4.4.3
Properties after immersion for 70 ± 1/2 hours in boiling water:	----	4.4.5
Volume change, percent, maximum (no shrinkage allowed)	15	4.4.5.1
Extraction, percent, maximum	0.5	4.4.5.2
Hardness change, durometer points, maximum	5	4.4.5.3

3.4 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Lot. - For the purposes of sampling, examinations, and tests, a lot shall consist of all valve disks of the same size, produced in one plant under essentially the same conditions, not exceeding 2500 pounds, and offered for delivery at one time.

4.2.2 Sampling for examination. - A random sample of valve disks shall be taken from each lot in accordance with the sampling plan given in table III for the examination specified in 4.3.1.

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Table III - Sampling for examination.

Number of valve ducts in lot	Number of valve ducts in sample	Number of nonconforming or defective valve disks			
		Major defects[1/]		Total defect[1/]	
		Acceptance number	Rejection number	Acceptance number	Rejection number
0 to 3	A11	0	0	1	2
4 to 15	3	0	1	1	2
16 to 50	8	0	1	1	2
51 to 90	13	1	1	1	2
91 to 150	20	1	2	2	3
151 to 280	32	1	2	3	4
281 to 500	50	2	3	4	5
501 to 1200	80	3	4	6	7
1201 and over	125	4	5	8	9

[1/] A major defect is a defect that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit. Total defects are major and minor defects combined.

4.2.3 Sampling for tests. - Random samples shall be taken from each lot that passes the examination of 4.3.1, in sufficient quantity to conduct either the production check tests or the quality conformance tests specified in 4.3.2 or 4.3.3, as applicable. Since the items are of such size and shape that all test specimens cannot be prepared from them, substitute samples shall be furnished with each lot in the form of three sample pieces 6 inches by 6 inches by 0.080 ± 0.010 inch thick and one sample piece approximately 4 inches by 3 inches by 0.500 ± 0.010 inch thick. The substitute samples shall be certified to be of the same material and equivalent cure as that used in the lot of finished material offered for delivery.

4.3 Quality conformance examination and tests. -

4.3.1 Examination. — Each of the samples selected in accordance with 4.2.2 shall be examined for workmanship, dimensions, tolerances, visual defects and any other requirement of this specification not covered by tests. MIL-STD-407 shall be used to determine and evaluate visual defects.

4.3.1.1 Rejection. - All defective or nonconforming disks shall be rejected and if the number of defective or nonconforming valve disks exceeds the applicable acceptance number of table III, the entire lot represented by the sample shall be rejected.

4.3.2 Production check tests. - Production check tests shall be conducted on samples from (or representing) the first lot of material and from every fifth lot thereafter. All the tests specified in 4.4 shall be conducted.

4.3.3 Quality conformance tests. - Quality conformance tests shall be conducted on samples from (or representing) all intermediate lots on which production check tests are not performed. The tests specified in 4.4.2, 4.4.3, and 4.4.4.3 shall be conducted.

4.3.4 Action in case of nonconformance. - If any of the samples in the production check tests or quality conformance tests is found not to be in conformance with requirements of this specification, the lot which it represents shall be rejected. Furthermore, additional samples shall be taken from each subsequent lot and they shall be subjected to the test or tests wherein the failure occurred. Each lot shall be accepted only after satisfactory results are obtained on the test or tests by all the samples taken to represent the lot. This additional testing shall be discontinued after four successive lots have passed the test or tests.

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4.4 Test methods.-

4.4.1 Testing conditions.- Unless otherwise specified herein, the testing conditions shall be in accordance with section 7 titled "Temperature and Methods of Conditioning Room" of FED-STD-601.

4.4.2 Tensile strength and ultimate elongation.- The tensile strength and ultimate elongation shall be determined by methods 4111 and 4121, respectively, of FED-STD-601. Die III specimens shall be used for all tensile determinations.

4.4.3 Hardness.- Hardness shall be determined in accordance with method 3021 of FED-STD-601 using a Shore A durometer.

4.4.4 Oven aging.- Method 7221 of FED-STD-601 shall be used for oven aging the test specimens as specified for the individual tests for 70 ± 1/2 hours at 100° ± 1.1°C (212° ± 2°F).

4.4.4.1 Tensile strength and elongation after oven aging.- The tensile strength and elongation after oven aging shall be determined by the procedure specified in 4.4.2.

4.4.4.2 Hardness after oven aging.- The hardness after oven aging shall be determined by the procedure specified in 4.4.3.

4.4.4.3 Compression set after oven aging.- Compression set shall be determined in accordance with method 3311 of FED-STD-601, with specimens clamped to 20 percent deflection and aged in accordance with 4.4.4.

4.4.5 Immersion in boiling water.- Specimens appropriate for the individual tests, prepared from the valve disks or samples provided in 4.2.3, shall be immersed in boiling 100° ± 1.1°C. (212° ± 2°F) distilled water for 70 ± 1/2 hours.

4.4.5.1 Volume change.- Volume change after immersion in boiling water shall be determined in accordance with method 6211 of FED-STD-601.

4.4.5.2 Extraction.- Extraction after immersion in boiling water shall be determined in accordance with method 6621 of FED-STD-601.

4.4.5.3 Hardness after immersion in boiling water.- The hardness of triplicate specimens shall be determined as specified in 4.4.3 before and after immersing the specimens in boiling water for 70 ± 1/2 hours. After the immersion period the specimens shall be conditioned for 16 to 96 hours at room temperature before testing.

4.5 Inspection of preparation for delivery.- A sample pack from each lot shall be examined for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early use and for storage of onboard repair parts.-

5.1.1 Packaging.- Packaging which may be the supplier's commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

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5.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination and date of cure.

5.1.4 Onboard repair parts. -

5.1.4.1 Packaging. - Unless otherwise specified, onboard repair parts shall be packaged by method III of MIL-P-116. Unit and intermediate containers shall conform to either PPP-B-566, PPP-B-636 or PPP-B-676 at the option of the supplier. Quantities per unit and intermediate containers shall be as specified (see 6.1). Closure and sealing of containers shall conform to the applicable container specification and appendix thereto.

5.1.4.2 Packing. - Onboard repair parts packaged as specified in 5.1.4.1 shall be packed as specified in 5.1.2.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Packaging. - Packaging shall be level A or C as specified (see 6.1).

5.2.1.1.1 Level A. - Unless otherwise specified, the valve disks shall be packaged by method III of MIL-P-116. Unit and intermediate containers shall conform to either PPP-B-566, PPP-B-636 or PPP-B-676 at the option of the supplier. Quantities per unit and intermediate containers shall be as specified. Closure and sealing of containers shall conform to the applicable container specification and appendix thereto.

5.2.1.1.2 Level C. - Packaging of valve disks shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from supply source to the using activity and until early installation. Packaging may conform to the supplier's commercial practice when such meets these requirements.

5.2.1.2 Packing. - Packing shall be level A, B or C as specified (see 6.1).

5.2.1.2.1 Level A. - Valve disks packaged as specified shall be packed for shipment in containers conforming to PPP-B-636, class weather resistant or PPP-B-640, class 2 at the option of the supplier. All corners and edge seams and manufacturer's joints of boxes shall be waterproofed with tape in accordance with the appendix to the box specification. Boxes shall be closed and banded with pressure sensitive tape in accordance with the applicable box specification or appendix thereto. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable fiberboard box specification. Intermediate fiberboard containers (see 5.2.1.1.1) conforming to weather resistant class of PPP-B-636, closed, sealed, and banded as specified herein may be used as the shipping container and need not be overpacked.

5.2.1.2.2 Level B. - Valve disks packaged as specified shall be packed for shipment in containers conforming to PPP-B-636, class domestic or PPP-B-640, Class 1 at the option of the supplier. Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of the containers shall not exceed the weight limitations of the applicable box specification. Intermediate containers (see 5.2.1.1.1) conforming to PPP-B-636, closed, sealed, and banded as specified herein may be used as the shipping container and need not be overpacked.

5.2.1.2.3 Level C. - The valve disks packaged as specified (see 6.1) shall be packed in a manner which will insure acceptance and safe delivery at destination. Containers shall comply with the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may be the supplier's commercial practice when such meets the requirements of this level.

5.2.1.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior containers shall be marked in accordance with MIL-STD-129 and date of cure.)

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of valve disk required (see 3.1).
(c) Quantity of valve disks per container (see 5.1.4.1 and 5.2.1.1.1).
(d) Packaging, packing and marking requirements, if other than 5.1 (see 5.2).

6.2 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 4820-N145SH)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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